

Collision-Free Tomato Harvesting with a 6-DOF Robotic Manipulator and Depth-Guided Motion Planning

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Abstract— *The increasing global demand for food production, coupled with persistent labour shortages in agriculture, has accelerated the need for intelligent robotic harvesting systems. Tomato harvesting presents unique challenges due to the dense foliage, delicate fruit structure, and unstructured environments found in greenhouse cultivation. In this study, we present a fully autonomous tomato harvesting system integrating real-time perception, path planning, and precision manipulation. The system employs a depth camera-based perception module to detect ripe tomatoes and surrounding obstacles, followed by a goal-biased Rapidly Exploring Random Tree (RRT) algorithm with path smoothing to generate collision-free trajectories for a 6-DOF serial manipulator. A custom-designed stem-cutting end-effector ensures efficient harvesting with minimal fruit damage. Kinematic analyses are performed using MATLAB Simulink, while real-time control and motion execution are handled on an NVIDIA Jetson Orin platform using the Robot Operating System (ROS). Experimental results demonstrate the effectiveness of the proposed approach in navigating cluttered environments and harvesting tomatoes with high success rates, highlighting the system's potential for practical application in smart agriculture systems.*

Keywords— *Autonomous Harvesting, Tomato Detection, Obstacle Avoidance, Goal-Biased RRT, 6-DOF Manipulator, Stem-Cutting End-Effector, Robot Operating System (ROS)*

I. INTRODUCTION

The agricultural sector continues to serve as a foundational component of the global economy, providing essential food supplies to support an ever-growing population. Within this context, tomato cultivation holds particular importance. In 2022, the European Union dedicated over 3.7 million hectares to fruit production and 2.0 million hectares to vegetables, with tomatoes accounting for more than 15 million tonnes of output [1].

Despite increasing global demand, agricultural productivity faces a significant threat: labour shortages. Recent Eurostat labour statistics highlight a persistent “lack of labour” in agriculture, primarily due to urban migration and an aging rural workforce [2]. This shortage is especially critical during harvest periods and has spurred increased interest in autonomous robotic systems.

Several robotic solutions have been introduced across different crop types. SCARA-type robotic arms have demonstrated high-speed and repetitive precision in cucumber harvesting under greenhouse conditions [3]. Similarly, kiwifruit harvesting robots have successfully integrated multi-arm systems with deep learning-based vision models [4]. Autonomous mobile platforms have also

emerged for field-level tasks, offering features like mapping, selective spraying, and navigation [5].

In the case of tomato harvesting, early vision-based systems primarily relied on structured-light sensors and HSV-based thresholding to detect ripe fruits [6]. Although this approach improved resilience to illumination variations, it often failed to generalize across field conditions. More recent developments have employed binocular stereo vision with Circular Hough Transform and RGB colour segmentation to better detect tomatoes in structured settings [7].

Mechanical aspects of harvesting have also evolved. Traditional grippers frequently damaged fruits, prompting researchers to explore alternatives. Stem-cutting end-effectors have demonstrated improved post-harvest quality [8], while recent systems like “AHPPEBot” combine phenotyping, pose estimation, and specialized key point detection to harvest tomatoes efficiently and selectively in complex clusters [9]. Another notable approach is the use of pneumatically actuated finger-like grippers, which offer compliance and minimal pressure distribution, enhancing delicate fruit handling during harvest [10].

Yet, physical interaction alone does not address the complexity of dense greenhouse environments, where stems, branches, and unripe fruits often obstruct access. Most existing systems lack the intelligence for real-time, adaptive obstacle avoidance. To tackle this, Lin et al. proposed a deep reinforcement learning-based planner using recurrent DDPG, which demonstrated over 90% success in simulating collision-free paths in highly cluttered orchards [11].

Considering these challenges, we present an autonomous tomato harvesting system that integrates real-time depth-based perception, adaptive motion planning, and a 6-DOF serial robotic manipulator equipped with a stem-cutting end-effector. “A goal biased Rapidly Exploring Random Tree” (RRT) algorithm with path smoothing is used for trajectory generation. The system operates on an NVIDIA Jetson Orin platform using the Robot Operating System (ROS) and has been validated both in simulation and physical tests.

Our approach addresses key gaps in agricultural robotics by combining perception and obstacle-aware planning with a cost-effective design made from 3D-printed parts. The system is scalable, well-suited for dense greenhouse environments, and contributes to the next generation of smart harvesting technologies [12].

II. MATERIALS AND METHODS

A. Kinematic structure of robot

The tomato-harvesting robotic system consists of a six-degree-of-freedom (6-DOF) serial manipulator, a custom-designed end-effector for grasping tomatoes, and a fixed base platform. The manipulator features six revolute joints, which provide sufficient dexterity to perform precise movements within the complex and confined layouts typically found in greenhouse tomato cultivation (Fig. 1). The end-effector is a simple parallel jaw gripper with a compact actuation mechanism, designed to securely grip tomatoes while minimizing fruit damage. It features symmetrical fingers and a cam-actuated linkage system that ensures consistent closing force and controlled motion, making it suitable for selective harvesting in dense foliage. A CAD representation of the gripper is shown in Fig. 2

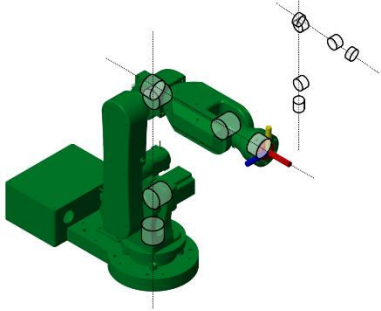


Fig. 1. Kinematic Structure Represented with CAD Model

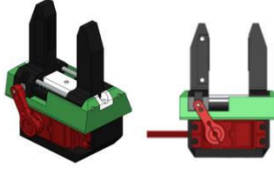


Fig. 2. CAD Representation of the Custom Parallel Jaw Gripper Designed for Tomato Harvesting

To model the kinematic structure of the manipulator, the standard Denavit–Hartenberg (D-H) convention was employed. Each link and joint in the manipulator were systematically assigned a coordinate frame based on this convention, enabling both forward and inverse kinematic analyses. The robot's structural layout, along with the associated link lengths l_1, l_2, l_3, l_4 and vertical offset a , is illustrated in Fig.3, representing the side view of the robot annotated with dimensional definitions.

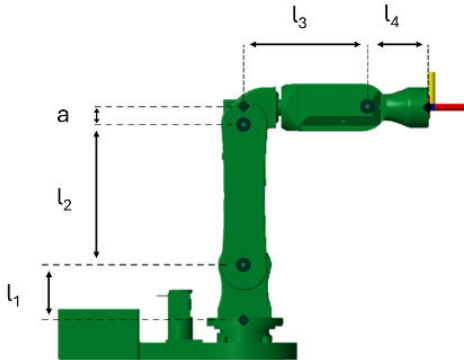


Fig. 3. Denavit-Hartenberg (D-H) Dimensions Assignments Represented with CAD Model

The complete D-H frame assignment is shown in Fig. 4, where the orientation of each joint coordinate system is defined following the right-hand rule. In this schematic, z_i axes are aligned with joint rotation axes and x_i axes represent the common normal between adjacent z -axes. This configuration serves as the foundation for deriving homogeneous transformation matrices used in forward kinematics.

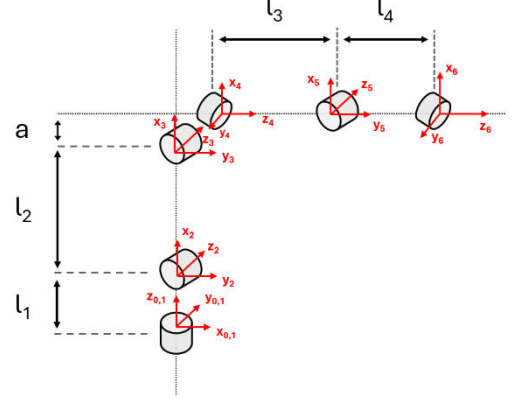


Fig. 4. Denavit-Hartenberg (D-H) Frame Assignments

Table 1 summarizes the D-H parameters for each joint i , including twist angles α_{i-1} , link lengths a_{i-1} , link offsets d_i , and joint variables θ_i . These parameters enable precise transformation from joint space to Cartesian space.

TABLE I. D-H Parameters of the Robot

Joint i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	$-\frac{\pi}{2}$	0	0	$-\frac{\pi}{2} + \theta_2$
3	0	l_2	0	θ_3
4	$-\frac{\pi}{2}$	a	l_3	θ_4
5	$\frac{\pi}{2}$	0	0	θ_5
6	$-\frac{\pi}{2}$	0	l_4	θ_6

Transformation matrices to compute the Cartesian position and orientation of the end-effector from a given set of joint angles. For inverse kinematics, MATLAB Simulink was utilized to compute the required joint configurations that allow the end-effector to reach target positions generated by the motion planning algorithm. This structure ensures reliable motion execution based on planned paths.

B. Simulation Environment and Kinematic Validation

A simulation environment was developed in MATLAB Simulink, using Simscape Multibody for accurate 3D dynamics modeling, to validate the manipulator's kinematic behavior and test motion planning strategies in greenhouse tomato harvesting scenarios. The environment models the 6-DOF manipulator, joint actuation mechanisms, and spatial constraints relevant to tomato harvesting in greenhouse conditions. Forward and inverse kinematics were implemented to ensure reachability and joint feasibility for target end-effector positions. Various workspace configurations and joint trajectories were tested

to confirm the manipulator's ability to navigate within complex environments without self-collision or singularity issues.

The simulation environment also served as a platform to pre-evaluate planned trajectories generated by the path planning algorithm. Joint angle profiles and end-effector paths were visualized and analyzed, supporting early-stage refinement before real-world deployment.

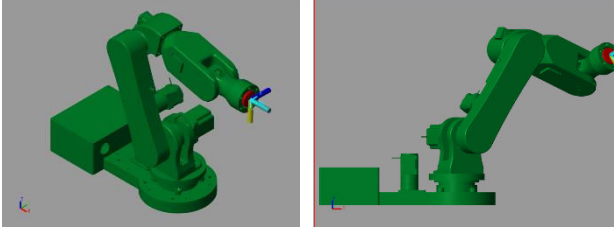


Fig. 5. Simscape Model View of the Manipulator at Initial and Target Configurations

A simulation environment was constructed in Simulink/Simscape to visualize the manipulator's kinematic reachability and joint coordination. Fig.5 illustrates the starting and target poses of the manipulator, while the corresponding modelling and communication structure are shown in Fig. 6

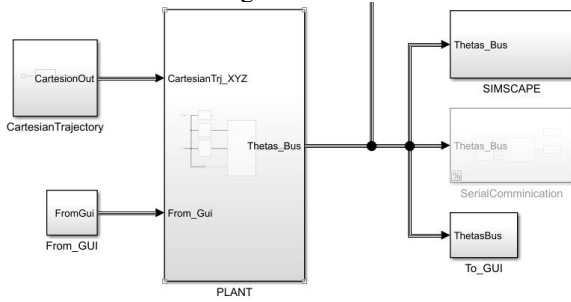


Fig. 6. Simulink block diagram of the simulation and Joint Control Architecture.

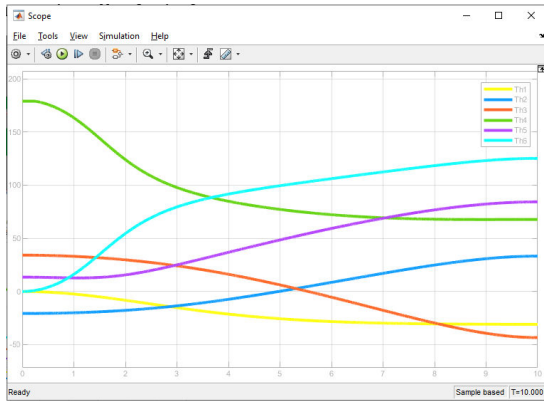


Fig. 7. Joint Angle Trajectories ($\theta_1-\theta_6$) Generated During the Simulated Motion

Fig. 7 shows the joint angle profiles generated during a simulated path execution from initial to target pose. The curves validate the feasibility and smoothness of the trajectory under real-time control constraints.

III. CONTROL AND PERCEPTION SYSTEM

A. Low-Level Joint Control

The finalized smoothed paths, represented as sequences of cartesian waypoints, were provided to the inverse kinematics module to generate joint-space trajectories. These joint commands were transmitted via CAN bus to STM32-based motor control boards, where each actuator was regulated by a closed-loop PID controller with integrated motor and encoder feedback for accurate joint tracking.



Fig. 8. Overview of the Robot in an Experimental Scene

All high-level processes, including RGB-D image acquisition, object detection, keypoint estimation, motion planning, and control, were managed by ROS 2 nodes running on a Jetson Orin. YOLOv8 inference was performed with CUDA acceleration, and the resulting target positions and obstacle data were used directly in the control pipeline.

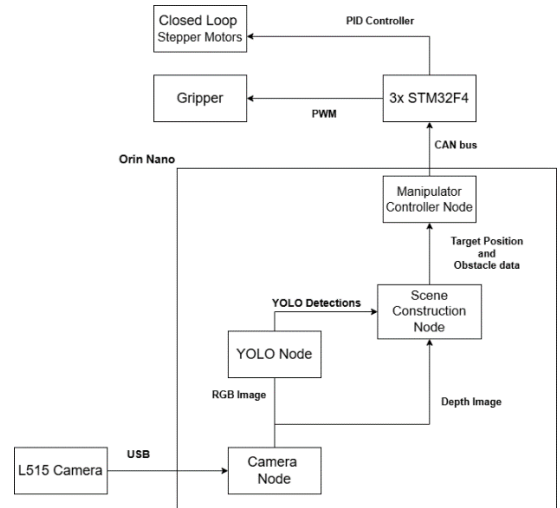


Fig. 9. System Diagram

Alongside the real-world implementation, the complete planning and control framework was also validated in a MATLAB-based simulation environment, where virtual scenarios with varying obstacle densities were used to test the system's reliability prior to deployment.

B. Vision System

To accurately reconstruct the scene for reliable perception and motion planning, an Intel RealSense L515 LiDAR camera was employed in this study. The sensor was configured to capture depth images at 620×420 resolution and RGB images at 1280×720 , both at 30 frames per second. Thanks to its time-of-flight (ToF) technology, the L515 was able to provide high-quality and detailed depth data, which is essential for precise three-dimensional understanding in indoor agricultural environments.

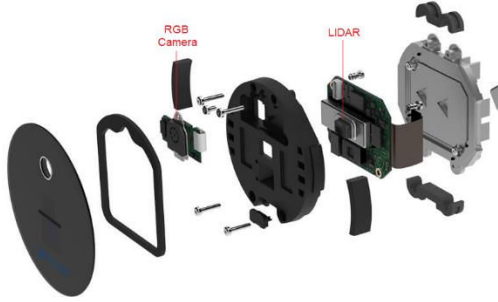


Fig. 10. Internal Structure of the L515 Camera

Although the L515's performance may be affected under strong ambient light or on reflective surfaces, it consistently produced accurate and reliable depth measurements within the conditions of the experimental setup. These measurements played a key role in enabling robust target localization and supporting the overall performance of the harvesting pipeline.



Fig. 11. Camera Placement

The RealSense camera was mounted on the fourth link of the robotic arm, forming a configuration that lies between conventional eye-to-hand and eye-in-hand setups. This hybrid placement was chosen to ensure a wide field of view prior to the onset of manipulator motion, facilitating comprehensive perception of both the target fruits and nearby obstacles. The RGB and depth data streams captured by the camera were processed in real-time on the Orin Nano.

C. Image processing and object localization

Accurate localization of ripe tomatoes and obstacles (such as plant branches or unripe fruits) is critical for effective path planning. Ripe, half-ripe, and unripe tomatoes were detected using a YOLOv8-based convolutional neural network (CNN) specifically trained for this task.

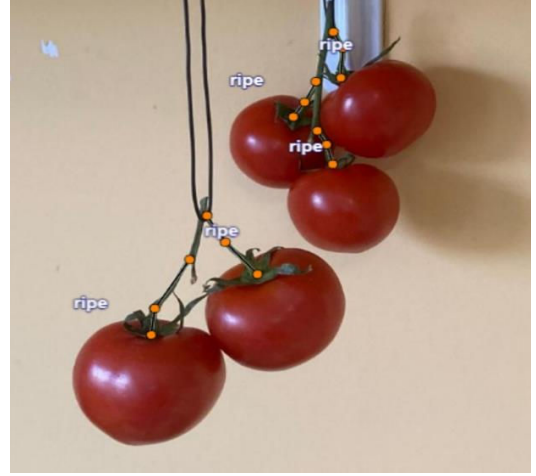


Fig. 12. Tomato Body and Stem Keypoint Detection

For ripe fruits, the model also predicted keypoints corresponding to the stem attachment point. The depth value at the detected stem keypoint was sampled from the aligned depth image, and using the pinhole camera model the corresponding three-dimensional location of the stem relative to the camera was computed. After transforming the 3D keypoint location to the robot base frame, it was used as the target input for the inverse kinematics module to enable motion planning toward the ripe tomato.

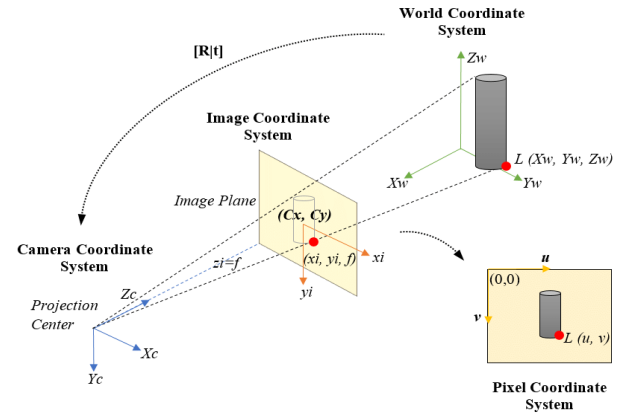


Fig. 13. Pinhole Camera Model [13]

To ensure safe motion planning toward the detected target, a subset of the point cloud was extracted based on its relative position by selecting all points located within a range starting 20 cm from the camera and extending up to 10 cm beyond the depth of the target key point. This localized point cloud, which contains potential obstacles such as unripe tomatoes, branches, and structural elements, was first processed with a statistical outlier removal filter that eliminates isolated or noisy points and then downsampled using a voxel grid filter to reduce point density and improve efficiency.

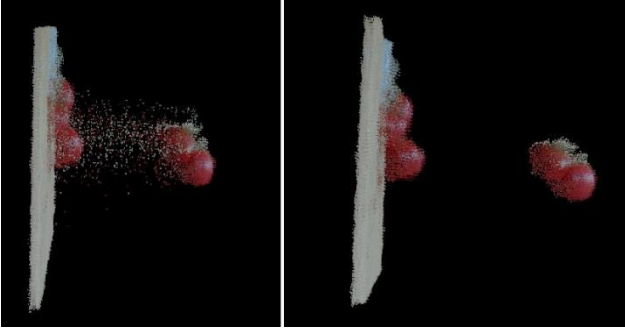


Fig. 14. Before and After Filtering the Point Cloud

Then, RANSAC-based model fitting was applied to isolate individual obstacle candidates and approximate them as simple geometric primitives (e.g., spheres, cylinders, boxes).

From each fitted shape, key spatial parameters such as centroid position, orientation, and bounding dimensions were extracted and utilized by the obstacle avoidance algorithm to construct collision objects and generate safe, collision-free motion trajectories toward the target.

IV. OBSTACLE-AWARE PATH PLANNING

To ensure safe navigation of the robotic manipulator in dense and unstructured tomato plant environments, a dedicated obstacle-aware path planning framework was developed. The system accounts for physical structures such as stems, branches, and unripe tomatoes, which are frequently encountered during harvesting operations and may obstruct access to target fruits.

The motion planning module receives 3D spatial representations of obstacles extracted from the RGB-D data stream. Following preprocessing and segmentation, each obstacle is approximated using a simple geometric primitive. In this work:

- Unripe tomatoes are modelled as spheres, based on their approximately isotropic shape and uniform size.
- Tomato stems and branches are modelled as cylinders, given their elongated and cylindrical physical structure.

These primitives are parameterized by spatial attributes such as center points, endpoints, radius, and orientation vectors, and are passed to the path planner as collision objects.

An adaptive, goal biased Rapidly-exploring Random Tree (RRT) algorithm is employed to compute an initial trajectory from the robot's current configuration to the desired stem cutting point. The planner is executed in a 3D configuration space, incorporating both spatial bounds and obstacle constraints. During node expansion, all candidate positions and connecting segments are subjected to obstacle intersection checks using both point-to-obstacle and segment-to-obstacle collision routines.

To improve path quality, the raw trajectory is post-processed in two steps:

1. **Path simplification**, which removes unnecessary intermediate waypoints using shortcut-based collision checks.
2. **Cubic spline interpolation**, which produces a smooth and continuous path suitable for real-time inverse kinematics and joint control.

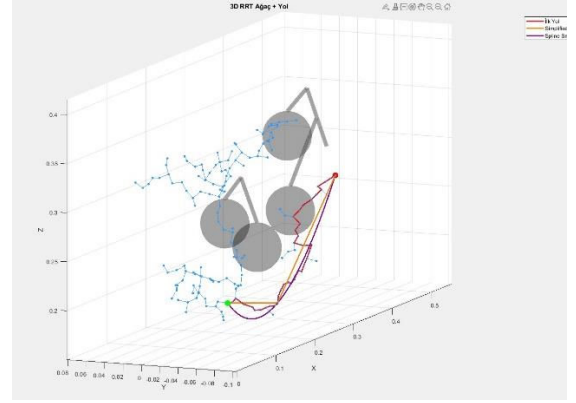


Fig. 15. Obstacle-Aware Path Planning with Geometrical Obstacles

The resulting path is then validated through pointwise collision checking to ensure that the smoothing process does not introduce any unintended intersection with obstacle regions. If a valid, collision-free spline cannot be found, the planning process is automatically re-invoked up to a predefined number of attempts.

This modular and geometry-aware planning strategy enables robust and reliable motion generation within the highly constrained environments of tomato cultivation.

V. RESULT AND DISCUSSION

The same scene configuration, with ripe tomatoes hanging between artificial branches in the laboratory, was replicated in the MATLAB simulation environment. The setup includes a 6-DOF robotic manipulator and simplified models of branches and tomatoes, placed using spatial data directly acquired from the real scene.

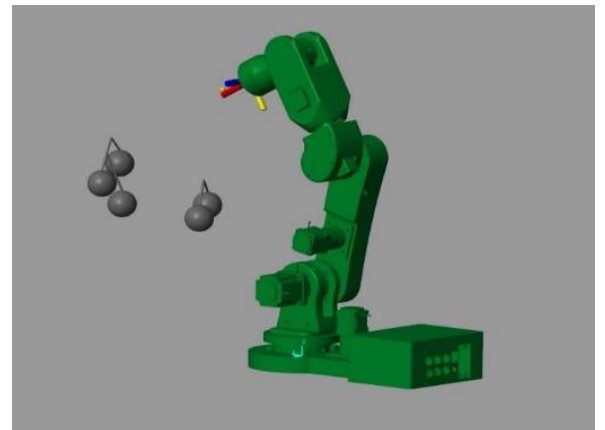


Fig. 16. Experimental Harvesting Scene in MATLAB Simscape

However, despite the effort to mimic the physical environment, simulation still relies on idealized obstacle shapes and deterministic object locations, which may not fully capture the complexity and uncertainty present in real-world agricultural scenarios. To narrow this gap, a downsampled point cloud representation of the actual scene can be integrated into the simulation to provide a more realistic and data-driven planning environment.

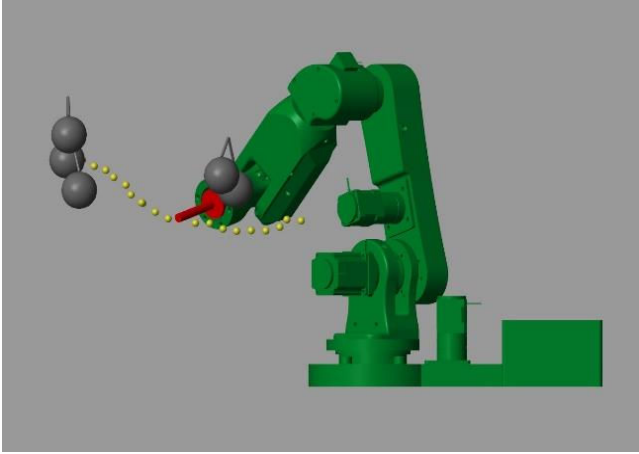


Fig. 17. Simulated Trajectory Planning Result

As illustrated in Fig. 18, the simulated manipulator follows the indicated trajectory and avoids obstacles successfully to reach the target fruit. While the goal-biased RRT provides promising results, an additional consideration involves the cubic spline interpolation used for path smoothing. This step introduces new intermediate points not included in the original planning phase, which may occasionally pass near or through obstacle regions. Although post-smoothing collision checks are applied, they may not always guarantee full safety in dense or irregular environments. Incorporating spline-aware planners or continuous collision monitoring during interpolation could further enhance robustness.



Fig. 18. Physical Trajectory Execution Result

The system exhibited consistent performance in both simulated and physical environments. Using the setup where real tomatoes were hung with artificial branches, the planned trajectory was successfully replicated on the physical robot. Although minor mechanical inaccuracies

(e.g., joint backlash) arise due to the 3D-printed structure, closed-loop motor control ensures that joint angles are accurately achieved, enabling reliable trajectory tracking. However, the inherent randomness of the RRT algorithm can limit repeatability across runs, which may affect results in physical deployments. Despite these challenges, the results confirm that the proposed framework is practically applicable and transferable to real-world scenarios under controlled conditions.

VI. ACKNOWLEDGMENT

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